B8 Statistical correctness

Survey weights, confidence intervals, effect sizes and multiple comparisons.

B8-L1 Survey weights and the effective sample size

Weighted estimates divide by the weight total. Precision uses the effective sample size.

```
weighted_mean = (y * w).sum() / w.sum()      # same as np.average(y, weights=w)
```

```
n_eff = w.sum() ** 2 / (w ** 2).sum()
design_effect = n / n_eff
```

Weighted mean	sum(y * w) / sum(w), over rows with both present. Same as np.average(y, weights=w).
Kish effective sample size	(sum w)^2 / sum(w^2). Design effect = n / n_eff.
Which n for intervals and suppression?	n_eff. Report the raw base as well.
Missing weight	Not zero. Drop the row from weighted estimates, and investigate.

B8-L2 Confidence intervals and effect sizes

Wilson intervals with n_eff for weighted shares, and an effect size beside every p-value.

```
from statsmodels.stats.proportion import proportion_confint
low, high = proportion_confint(count=40, nobs=100, alpha=0.05, method="wilson")   # (0.309, 0.498)
```

Why not the Wald interval?	Zero width at p = 0 or 1, and too narrow for small n. Use Wilson.
Interval for a weighted share	Wilson with the weighted p and n_eff.
Significant vs meaningful	Large samples make trivial differences significant. Report the size with its interval.
Cohen's h	`2 * asin(sqrt(p1)) - 2 * asin(sqrt(p2))`. About 0.2 small, 0.5 medium, 0.8 large.

B8-L3 Many tests, and the false positives they bring

Define the family first, adjust within it with Holm or Benjamini-Hochberg, and report how many tests ran.

```
from statsmodels.stats.multitest import multipletests
reject, p_adj, _, _ = multipletests(p_values, alpha=0.05, method="holm")
```

Chance of at least one false positive in 20 tests at 5%	1 - 0.95^20 = 0.64.
Holm vs Bonferroni	Same family-wise control. Holm is step-down and rejects at least as many. Prefer Holm.
Benjamini-Hochberg	Controls the false discovery rate. Suits screening many cells.
Define the family	Before looking at results. Report how many tests ran.